PHYS1408 Practice Exam 2-8 — Answer Key

Worked solutions

All answers were checked numerically with `_scripts/build_exam_2_8.py`. $g = 9.8$ m/s^2 throughout.

Practice Exam 2-8 is built from the professor's **lecture-note examples**, one per question, with new numbers. It keeps his setups and his unknowns: the halfpipe after the rain, the roller coaster, the four incline cases, the spring at the bottom of a ramp, the man on the stairs, the block that "doesn't go" up the incline, loop launch height, the 3×5-notecard elastic collision, the rod on the merry-go-round, the ballistic pendulum, the pulley "Atwood's machine w/ torque," the rolling disk, and the disk dropped on a disk. Question 9 is a **minimum** trap: round up. The last column says which 3×5 notecard entry solves each question.

Q	Answer	Value	Topic	Notecard entry
1	**a**	2.34 m	Halfpipe with a rough strip	Halfpipe
2	**b**	29.7 m/s	Roller coaster	Fall / coaster
3	**a**	2.95 m/s	Sliding up a frictionless incline	One block on incline (Up, $\mu = 0$)
4	**b**	1.49 m	Rough incline, sliding up: stopping distance	One block on incline (Stop up)
5	**c**	2.94 m	Rough incline, sliding down: stopping distance	One block on incline (Stop down)
6	**e**	0.20 m	Mass slides into a spring	Spring (stop a mass)
7	**b**	941 W	Power climbing stairs	Power
8	**e**	stops (at 0.77 m)	Incline + hanging mass, stops early	Incline + hanging mass + pulley
9	**d**	22.5 m → 24	Loop-the-loop, minimum launch height	Loop
10	**e**	+5.5 m/s	1-D elastic collision	Elastic
11	**c**	1.44 rad/s	Rod dropped from the center to the edge	Angular momentum (I_{add}: rod)
12	**d**	419 m/s	Ballistic pendulum, bullet speed	Ballistic
13	**d**	3.57 m/s	Hanging mass on a disk pulley	Atwood / mass on pulley (one mass)

Q	Answer	Value	Topic	Notecard entry
14	**c**	4.85 m/s	Disk rolling down an incline	Rolling (disk row)
15	**a**	2.20 rad/s	Disk dropped on a spinning disk	Angular momentum

1. Halfpipe with a rough strip — a) 2.34 m

Source: Ch 7/8 notes p. 4/18–5/18. **Notecard:** Halfpipe.

Notecard, from rest, one crossing:

$$h_f = h_i - \mu L$$

Plug in:

$$h_f = 2.5 - (0.02)(8.0) = 2.5 - 0.16 = 2.34 \text{ m}$$

Distractors:

- b)–e) subtract the friction loss 1.5, 2, 2.5 and 3 times. Crossing the strip once loses μL once.

2. Roller coaster — b) 29.7 m/s

Source: Ch 7/8 notes p. 3/18. **Notecard:** Fall / coaster.

Notecard (no friction, from rest):

$$v_f = \sqrt{v_i^2 + 2g(h_i - h_f)} = \sqrt{2g(h_i - h_f)}$$

Plug in:

$$v_f = \sqrt{2(9.8)(75 - 30)} = \sqrt{882} = 29.7 \text{ m/s}$$

Distractors:

- a) 21.0 forgets the 2.
- c) 38.3 uses the full 75 m drop.
- d) 42.0 doubles the drop.
- e) 882 forgets the square root.

3. Sliding up a frictionless incline — a) 2.95 m/s

Source: Ch 7/8 notes p. 6/18. **Notecard:** One block on incline (Up, $\mu = 0$).

Notecard, sliding up, with $\mu = 0$:

$$v_f = \sqrt{v_i^2 - 2g\,\Delta x \sin\theta}$$

Plug in ($\sin 12^\circ = 0.2079$):

$$v_f = \sqrt{5.0^2 - 2(9.8)(4.0)(0.2079)} = \sqrt{25 - 16.30} = \sqrt{8.70} = 2.95 \text{ m/s}$$

Distractors:

- b) 4.10 forgets the 2.
- c) 6.43 adds instead of subtracts.
- d) 8.70 forgets the square root.
- e) comes from using $\cos\theta$.

4. Rough incline, sliding up: stopping distance — b) 1.49 m

Source: Ch 7/8 notes p. 9/18. **Notecard:** One block on incline (Stop up).

Notecard, stop up:

$$\Delta x = \frac{v_i^2}{2g(\sin\theta + \mu\cos\theta)}$$

Plug in ($\sin 15^\circ = 0.2588$, $\cos 15^\circ = 0.9659$):

$$\Delta x = \frac{4.0^2}{2(9.8)(0.2588 + 0.30 \times 0.9659)} = \frac{16}{19.6(0.5486)} = 1.49 \text{ m}$$

Distractors:

- a) 0.74 has an extra factor of 2.
- c) 2.98 forgets the 2.
- d) 3.15 ignores friction.
- e) 6.30 ignores friction and the 2.

5. Rough incline, sliding down: stopping distance — c) 2.94 m

Source: Ch 7/8 notes p. 10/18. **Notecard:** One block on incline (Stop down).

It stops, because $\mu\cos\theta = 0.248$ is bigger than $\sin\theta = 0.139$. Notecard, stop down:

$$\Delta x = \frac{v_i^2}{2g(\mu\cos\theta - \sin\theta)}$$

Plug in ($\sin 8^\circ = 0.1392$, $\cos 8^\circ = 0.9903$):

$$\Delta x = \frac{2.5^2}{2(9.8)(0.25 \times 0.9903 - 0.1392)} = \frac{6.25}{19.6(0.1084)} = 2.94 \text{ m}$$

The bracket is small (0.108), so keep 4 digits in the trig values.

Distractors:

- a) 0.82 uses the sliding-**up** bracket.
- b) 1.29 leaves out $\sin\theta$.
- d) 5.88 forgets the 2.
- e) It does stop, since $\mu\cos\theta > \sin\theta$.

6. Mass slides into a spring — e) 0.20 m

Source: Ch 7/8 notes p. 11/18. **Notecard:** Spring (stop a mass).

Notecard, stop a mass, starting from rest ($v_i = 0$):

$$x = \sqrt{\frac{2mgh + mv_i^2}{k}} = \sqrt{\frac{2mgh}{k}}$$

Plug in:

$$x = \sqrt{\frac{2(2.0)(9.8)(1.5)}{1500}} = \sqrt{0.0392} = 0.20 \text{ m}$$

Distractors:

- a) 0.020 is mgh/k (no 2, no root).
- b) 0.039 forgets the square root.
- c) 0.099 has the 2 upside down.
- d) 0.14 forgets the 2.

7. Power climbing stairs — b) 941 W

Source: Ch 7/8 notes p. 14/18. **Notecard:** Power.

Notecard:

$$P = \frac{mgh}{t}$$

Plug in:

$$P = \frac{(80)(9.8)(18)}{15} = \frac{14{,}112}{15} = 941 \text{ W}$$

Distractors:

- a) 96 forgets g.
- c) 1880 doubles it.
- d) 14,100 is the work mgh in joules, not power.
- e) 254,000 multiplies by t.

8. Incline + hanging mass, stops early — e) it stops before it slides the whole 1.0 m

Source: Ch 7/8 notes p. 15/18–16/18. **Notecard:** Incline + hanging mass + pulley.

Notecard, m_2 falls (m_1 slides **up**), massless pulley:

$$v_f = \sqrt{v_i^2 + \frac{2g\,\Delta x\,B}{m_1 + m_2}}$$

$$B = m_2 - m_1 \sin\theta - \mu m_1 \cos\theta$$

Net pull:

$$B = 3.0 - 15(0.4226) - 0.30(15)(0.9063) = 3.0 - 6.339 - 4.078 = -7.42 \text{ kg}$$

Plug in:

$$v_f = \sqrt{2.5^2 + \frac{2(9.8)(1.0)(-7.42)}{18}} = \sqrt{6.25 - 8.08} = \sqrt{-1.83}$$

Negative under the root: it "doesn't go 1.0 m." **Where it stops** (notecard):

$$\Delta x = \frac{-(m_1 + m_2)v_i^2}{2gB} = \frac{-(18)(6.25)}{2(9.8)(-7.42)} = 0.77 \text{ m}$$

Distractors:

- a) 1.49 forgets the 2 in $2g$.
- b) 1.62 ignores friction.
- c) 2.33 uses the m_1-slides-down B.
- d) 3.79 adds the bracket instead of subtracting it.

9. Loop-the-loop, minimum launch height — d) 24 m

Source: Ch 7/8 notes p. 12/18–13/18. **Notecard:** Loop.

Notecard: launch $h = \frac{5}{2}R$ (from $v_t = \sqrt{gR}$ at the top plus energy).

Plug in:

$$h = \tfrac{5}{2}(9.0) = 22.5 \text{ m}$$

Answer-choice trap. 22.5 m is a **minimum**. From 22 m the car falls off at the top, even though 22 is the closest number. The smallest choice that works is **d) 24 m**.

Distractors:

- a) 18 m is $2R$: it reaches the top with zero speed.
- c) 22 m is the nearest number, but it's below the minimum.
- e) 27 m is $3R$ (works, but isn't the minimum choice).

10. 1-D elastic collision — e) +5.5 m/s

Source: Ch 9 notes p. 3/5. **Notecard:** Elastic.

Notecard formulas, right = +: $m_1 = 6.0$, $v_{1i} = +4.0$; $m_2 = 2.0$, $v_{2i} = +1.0$.

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2 v_{2i}}{m_1 + m_2} = \frac{(4.0)(4.0) + 2(2.0)(1.0)}{8.0} = \frac{20}{8} = +2.5 \text{ m/s}$$

$$v_{2f} = v_{1i} + v_{1f} - v_{2i} = 4.0 + 2.5 - 1.0 = +5.5 \text{ m/s}$$

Distractors:

- a) +1.0 is no change.
- b) +2.5 is the 6.0 kg ball's velocity.
- c) +3.25 is the stick-together answer.
- d) +4.0 assumes the balls swap velocities.

11. Rod dropped from the center to the edge — c) 1.44 rad/s

Source: Ch 11 notes p. 5/8. **Notecard:** Angular momentum (I_{add}: rod).

Notecard with I_{add} = rod center→edge = $\frac{1}{3}m\ell^2$:

$$\omega_f = \frac{I_1\omega_1}{I_1 + \frac{1}{3}m\ell^2}$$

Moments of inertia: $I_1 = \frac{1}{2}(300)(2.5)^2 = 937.5$; $\frac{1}{3}(50)(2.5)^2 = 104.2$ (kg · m²).

Plug in:

$$\omega_f = \frac{(937.5)(1.6)}{937.5 + 104.2} = \frac{1500}{1041.7} = 1.44 \text{ rad/s}$$

Distractors:

- a) 1.20 treats the rod as a point mass at the rim ($m\ell^2$).
- b) 1.37 uses $\frac{1}{2}m\ell^2$.
- d) 1.56 uses the rod-about-its-middle formula, $\frac{1}{12}m\ell^2$.
- e) 1.60 ignores the rod.

12. Ballistic pendulum, bullet speed — d) 419 m/s

Source: Ch 9 notes p. 5/5. **Notecard:** Ballistic.

Notecard:

$$v = \frac{M + m}{m}\sqrt{2gh}$$

Plug in:

$$v = \frac{2.006}{0.006}\sqrt{2(9.8)(0.080)} = 334.3 \times 1.252 = 419 \text{ m/s}$$

Distractors:

- a) 1.25 is $V = \sqrt{2gh}$, the block's speed after the hit.
- b) 22.9 uses energy conservation through the collision.
- c) 334 is just the mass ratio.
- e) 837 doubles it.

13. Hanging mass on a disk pulley — d) 3.57 m/s

Source: Ch 10 notes p. 4/8–6/8. **Notecard:** Atwood / mass on pulley (one mass).

Notecard, one mass: $v_f^2 = v_i^2 + \dfrac{2mgh}{m + I/R^2}$. A disk has $I/R^2 = \frac{1}{2}M = 3.0$ kg.

$$v_f = \sqrt{v_i^2 + \frac{2mgh}{m + \frac{1}{2}M}}$$

Plug in:

$$v_f = \sqrt{1.0^2 + \frac{2(2.0)(9.8)(1.5)}{2.0 + 3.0}} = \sqrt{1.0 + 11.76} = \sqrt{12.76} = 3.57 \text{ m/s}$$

Distractors:

- a) 2.62 forgets the 2.
- b) 2.89 treats the pulley as a ring ($I/R^2 = M$).
- c) 3.43 forgets v_i^2.
- e) 5.51 ignores the pulley (free fall).

14. Disk rolling down an incline — c) 4.85 m/s

Source: Ch 10 notes p. 7/8–8/8. **Notecard:** Rolling (disk row).

Notecard, disk row: $v = \sqrt{\frac{4}{3}gh}$.

Plug in:

$$v = \sqrt{\tfrac{4}{3}(9.8)(1.8)} = \sqrt{23.52} = 4.85 \text{ m/s}$$

Distractors:

- a) 3.43 uses $\frac{2}{3}$ instead of $\frac{4}{3}$.
- b) 4.20 is a ring, $\sqrt{gh}$.
- d) 5.02 is a sphere, $\sqrt{\frac{10}{7}gh}$.
- e) 5.94 is sliding, $\sqrt{2gh}$.

15. Disk dropped on a spinning disk — a) 2.20 rad/s

Source: Ch 11 notes p. 1/8. **Notecard:** Angular momentum.

Notecard with $\omega_2 = 0$:

$$\omega_f = \frac{I_1 \omega_1}{I_1 + I_2}$$

Moments of inertia: $I_1 = \frac{1}{2}(400)(2.0)^2 = 800$; $I_2 = \frac{1}{2}(150)(1.2)^2 = 108$ (kg $\cdot$ m^2).

Plug in:

$$\omega_f = \frac{(800)(2.5)}{800 + 108} = \frac{2000}{908} = 2.20 \text{ rad/s}$$

Distractors:

- b) 2.50 is no change.
- c) 2.84 flips the ratio.
- d) 4.40 doubles it.
- e) 5.00 doubles the starting speed.